ECA13 – MICROWAVE ENGINEERING

CO1 Application-Based Assessment

Analysis, Design and Verification of an X-Band Weather-Radar Waveguide Feed Network

Technical Report – Worked Answers

Selected application: X-band weather radar feed using a WR-90 rectangular waveguide operating near 10 GHz.

Design note: Numerical values are proposed academic targets and first-order calculations. Loss and matching figures identified as estimates must be verified by electromagnetic simulation and calibrated measurements.

Stage 1 – Formulate the Application and Operating Requirement

1. Application and importance

The selected application is the low-loss RF feed between the transmitter/receiver front end and the antenna of an X-band weather radar. A rectangular waveguide is suitable because it can carry microwave power with low conductor loss and good shielding. At 10 GHz, connector transitions, flange alignment, surface finish and dimensional tolerances affect reflections and insertion loss.

Microwave engineering developed from electromagnetic field theory and early experiments demonstrating radio waves. The cavity magnetron enabled high-power microwave generation for radar. Later advances in semiconductor devices, satellite links, radio astronomy and weather sensing broadened microwave applications. Modern weather radars use reflected echoes to estimate precipitation location, intensity and motion.

2. Compare frequency bands

Band	Example range	Free-space wavelength	Advantages	Limitations
S band	2–4 GHz	150–75 mm	Lower rain attenuation; good range	Larger antenna/feed
C band	4–8 GHz	75–37.5 mm	Coverage/bandwidth compromise	Moderate antenna size
X band	8–12 GHz	37.5–25 mm	Compact antenna; good spatial detail	Higher rain attenuation than S band

The selected operating band is 9.3–9.5 GHz, centred at 9.4 GHz, with a 200 MHz bandwidth. X-band supports compact radar hardware and fine spatial resolution, but rain attenuation and alignment sensitivity are more significant than at lower frequencies.

3. Quantified requirement specification

Parameter	Proposed requirement
Application	X-band weather radar antenna feed
Operating band	9.3–9.5 GHz; centre 9.4 GHz
Transmission structure	WR-90 rectangular waveguide
Internal dimensions	a = 22.86 mm; b = 10.16 mm
Feed length	1.5 m nominal example
Insertion-loss target	$\leq$ 0.5 dB for guide section; verify experimentally
Return loss	$\geq$ 20 dB at reference flange after tuning
Power	Set by transmitter and component ratings; verify thermal limits
Cost target	Academic prototype target $\leq$ ₹5,000 where locally available; excludes instruments

4. Constraints and acceptance criteria

The feed must maintain a continuous conducting boundary, correct flange alignment and secure electrical contact. Bends and transitions should be minimized. Acceptance requires operation throughout 9.3–9.5 GHz, return loss of at least 20 dB at the reference plane, and measured insertion loss within the agreed budget. These are project targets rather than regulatory limits.

Stage 2 – Evaluate and Select the Transmission Structure

5. Mode comparison

Structure	Mode	Strengths	Limitations for radar feed
Coaxial line	TEM	Flexible, shielded, convenient interfaces	Loss/power limits depend on cable/connectors
Rectangular waveguide	TE/TM; dominant TE10	Low microwave loss; good power handling	Bulkier; flange/transition alignment
Stripline	TEM	Shielding and compact routing	PCB loss and power handling need evaluation
Microstrip	Quasi-TEM	Compact and inexpensive	Radiation and dielectric loss at X-band
Coplanar waveguide	Quasi-TEM	Easy probing and planar integration	Ground geometry affects impedance

A hollow single-conductor waveguide cannot support TEM because TEM requires both electric and magnetic fields transverse to propagation and a two-conductor voltage-return arrangement. The dominant propagating mode in a standard rectangular guide is TE10 when operating above its cutoff and below competing-mode cutoffs.

6. Weighted decision matrix

Scores use 1 = poor and 5 = excellent for the selected 10 GHz radar feed. Weighted score = sum(weight × score)/100.

Criterion	Weight %	Coax	WR-90	Stripline	Microstrip
Low loss	25	3	5	3	2
Power handling	20	3	5	3	2
Compact size	10	4	2	5	5
Cost/availability	10	4	3	4	5
Shielding/EMI	15	4	5	5	2
Radar-feed suitability	20	3	5	2	2
Weighted score / 5	100	3.35	4.50	3.45	2.70

Selection: WR-90 is preferred for the main X-band feed because low loss, power handling and shielding are more important than minimum size. Coaxial cable remains useful for short instrument connections and transitions; planar lines may suit low-power board-level circuitry.

Design Decision 1

Select WR-90 rectangular waveguide for the principal radar feed. The choice is conditional on confirming flange compatibility, return loss and insertion loss using the actual hardware.

Stage 3 – Develop the Mode and Propagation Model

7. Waveguide equations

For a rectangular waveguide with broad dimension a and narrow dimension b, the cutoff frequency for TE_mn/TM_mn modes is:

$fc(mn) = (c/2)\sqrt{[(m/a)^2 + (n/b)^2]}$

For the dominant TE10 mode, n = 0 and m = 1:

$fc10 = c/(2a)$

Above cutoff, phase constant and guide wavelength are:

$\beta = \sqrt{(k_o^2 - kc^2)}$; $\lambda g = 2\pi/\beta = \lambda_o / \sqrt{[1 - (fc/f)^2]}$

For TE modes, wave impedance is $ZTE = \omega\mu/\beta$. Phase velocity $vp = \omega/\beta$ and group velocity $vg = c^2/vp$ in an ideal air-filled guide.

8. Waveguide dimensions and cutoff frequencies

Use WR-90 nominal internal dimensions a = 22.86 mm and b = 10.16 mm, with $c = 3.00 \times 10^8$ m/s.

$fc10 = 3.00 \times 10^8 / [2(22.86 \times 10^{-3})] \approx 6.56$ GHz

$fc20 = c/a \approx 13.11$ GHz

$fc01 = c/(2b) \approx 14.76$ GHz

At 9.4 GHz, TE10 is above cutoff while TE20 and TE01 are below cutoff. Thus the guide supports the intended dominant mode without these higher-order modes propagating in an ideal straight section.

9. Propagation parameters at 9.4 GHz

$\lambda_o = c/f \approx 31.91$ mm

$\lambda g = \lambda_o / \sqrt{[1 - (6.56/9.4)^2]} \approx 44.5$ mm

$\beta = 2\pi/\lambda g \approx 141$ rad/m

$vp \approx 4.18 \times 10^8$ m/s

$vg \approx 2.15 \times 10^8$ m/s

$ZTE \approx 526\ \Omega$

For a 1.5 m section, approximate group delay is $\tau g = L/vg \approx 1.5/(2.15 \times 10^8) \approx 6.98$ ns. These are ideal results; transitions, bends and tolerances affect actual behaviour.

10. TEM, TE and TM comparison

Mode	Field/cutoff property	Dispersion / impedance	Typical use
TEM	Ez = 0, Hz = 0; no cutoff in two-conductor line	Ideal uniform medium non-dispersive	Coaxial cable, stripline
TE	Ez = 0; Hz generally nonzero; cutoff in guide	Dispersive; $ZTE = \omega\mu/\beta$	Rectangular waveguide
TM	Hz = 0; Ez generally nonzero; cutoff in guide	Dispersive; $ZTM = \beta/(\omega\varepsilon)$	Waveguide modes
Quasi-TEM	Mostly transverse fields	Approximate impedance and effective permittivity	Microstrip/CPW

Design Decision 2

The dominant mode is TE10. At 9.4 GHz, calculated guide wavelength is approximately 44.5 mm and TE wave impedance approximately 526 Ω. Final design should confirm band, mode purity and transitions with simulation or measurement.

Stage 4 – Evaluate Losses and Design a Loss-Reduction Strategy

11. Loss mechanisms

Total feed loss can include conductor loss, flange/contact loss, mismatch loss, bend loss and transition loss. Actual values depend on conductivity, internal surface finish, alignment, flange condition and geometry. The following is an illustrative loss budget for a 1.5 m guide, not a measured WR-90 result.

Loss component	Initial estimate (dB)	Improved target (dB)	Share	Trade-off / remarks
Guide-wall conductor loss	0.20	0.12	44%	Better surface finish; higher cost
Flange/contact loss	0.10	0.05	22%	Clean and align mating faces
Bends/transitions	0.08	0.04	18%	Use well-designed transitions
Mismatch allowance	0.07	0.04	16%	Tune interfaces and verify return loss
Total illustrative loss	0.45	0.25	100%	1.5 m example; confirm by test

The illustrative budget reduces loss from 0.45 dB to 0.25 dB, an improvement of 0.20 dB. Actual attenuation must be established from manufacturer data, electromagnetic analysis or calibrated insertion-loss measurement.

12. Worked conductor-loss considerations

For a good conductor, skin depth and surface resistance are approximately:

$$\delta = \sqrt{[2/(\omega\mu_0\sigma)]} \; ; \quad R_s = \sqrt{(\pi f \mu_0/\sigma)}$$

For copper at 9.4 GHz using $\sigma \approx 5.8 \times 10^7$ S/m, skin depth is about 0.68 μm. Surface currents are concentrated near the conducting wall, so surface roughness and plating quality can increase attenuation. Actual guide material and finish should be used for a defensible calculation.

13. Parametric trends and redesign

Frequency: conductor surface resistance generally increases approximately with √f; propagation and cutoff relationships also change.

Surface finish: roughness comparable to skin depth can increase wall loss; specify a suitable smooth conducting surface.

Flanges: dirt, corrosion, gaps or poor alignment can increase mismatch and leakage.

Geometry: avoid unnecessary bends and abrupt transitions; use properly designed bends and twists.

Verification: measure insertion loss and return loss over 9.3–9.5 GHz with calibrated equipment and compare against the ideal model.

Design Decision 3

The illustrative largest contribution is waveguide-wall conductor loss. Improved surface finish, accurate flange mating and reduced transition discontinuities are the principal measures. The estimated loss target is 0.25 dB over 1.5 m, subject to experimental verification.

Stage 5 – Create and Evaluate the Network Model

14. Equivalent voltage/current and wave variables

In a hollow waveguide, unique voltage and current definitions are not generally available as they are for a TEM line. A modal equivalent circuit can nevertheless be used with a consistent mode normalization and wave impedance. For a two-port measurement, incident and reflected waves are normalized to the measurement system.

a_n = incident wave at port n; b_n = reflected/outgoing wave at port n

Scattering parameters describe how incident waves at the ports produce outgoing waves. S11 is input reflection, S21 forward transmission, S12 reverse transmission and S22 output reflection.

15. ABCD model and S-parameters

For a uniform transmission-line section with characteristic impedance Zc, propagation constant γ and length l:

$M = [[\cosh(\gamma l), Z_c \sinh(\gamma l)], [\sinh(\gamma l)/Z_c, \cosh(\gamma l)]]$

For cascaded sections, multiply matrices in physical order: Mtotal = M1 × M2. For ABCD parameters referenced to equal real impedance Z0:

$\Delta = A + B/Z_0 + CZ_0 + D$

$S11 = (A + B/Z_0 - CZ_0 - D)/\Delta$

$S21 = 2/\Delta$; $S22 = (-A + B/Z_0 - CZ_0 + D)/\Delta$

$S12 = 2(AD - BC)/\Delta$

A passive reciprocal network should satisfy S12 ≈ S21 at corresponding reference planes. For a waveguide, modal reference impedance and calibration planes must be specified; a simple 50 Ω microstrip interpretation should not be applied blindly.

16. Load mismatch example

For an illustrative equivalent interface with reference impedance Z0 = 50 Ω and load ZL = 60 Ω:

$\Gamma = (ZL - Z_0)/(ZL + Z_0) = 10/110 = 0.0909$

$VSWR = (1 + |\Gamma|)/(1 - |\Gamma|) \approx 1.20$

Return loss $= -20 \log_{10}|\Gamma| \approx 20.83$ dB

Mismatch loss $= -10 \log_{10}(1 - |\Gamma|^2) \approx 0.036$ dB

This equivalent example meets a 20 dB return-loss target narrowly. It illustrates mismatch calculations; it is not a claim that the waveguide flange itself has a 50 Ω or 60 Ω lumped impedance.

17. Quarter-wave matching concept

A quarter-wave transformer is common for TEM or quasi-TEM lines. For a real 60 Ω load on a 50 Ω line:

$ZT = \sqrt{(Z_0 ZL)} = \sqrt{(50 \times 60)} \approx 54.77\ \Omega$

This is included as a transmission-line comparison, not as a direct physical WR-90 matching prescription. A waveguide discontinuity or transition must be designed using modal analysis or a full-wave solver.

18. Network summary

Parameter	Initial / example	Improved target	Interpretation
Operating band	9.3–9.5 GHz	9.3–9.5 GHz	Verify over full band
Dominant mode	TE10 intended	TE10 maintained	Avoid mode conversion
Return loss	≥20 dB target	≥20 dB target	Measure at defined flange planes
Insertion loss	≤0.5 dB target	As low as practical	Measure after calibration
Reciprocity	Expected	S12 ≈ S21	Check measurement consistency

No measured or simulated S-parameter data were supplied. Numerical network performance remains a target and must be demonstrated by a calibrated vector network analyzer or validated electromagnetic model.

Stage 6 – Validate, Critique and Defend the Final Design

19. Specification verification

Requirement	Specified	Initial estimate	Verification / status
Operating band	9.3–9.5 GHz	Selected band	Measure across band; pending
Mode	TE10 dominant	Cutoff supports TE10	Confirm transitions; provisional
Insertion loss	≤0.5 dB target	0.45 dB illustrative	Estimate only; measure
Return loss	≥20 dB target	Not measured	VNA measurement required
Feed length	1.5 m nominal	1.5 m assumed	Confirm physical assembly
Power	Per transmitter rating	Not calculated	Check thermal/component ratings

20. Tolerance analysis

Waveguide dimensions and flange alignment affect cutoff frequency, phase and reflection. From fc10 = c/(2a), increasing a lowers cutoff, while decreasing a raises it. A small gap or lateral misalignment at a flange can create a discontinuity even if nominal guide dimensions are correct. Use manufacturer tolerances and evaluate worst-case dimensional and alignment conditions.

21. Assumptions and their effects

- Ideal conducting walls neglect finite conductivity and surface roughness, so conductor loss is underestimated.
- Air-filled uniform guide neglects moisture, contamination and dielectric loading.
- Straight uniform guide ignores bends, twists, flanges and transitions.
- Single-mode calculation assumes higher-order modes are not excited by discontinuities.
- Nominal dimensions do not include manufacturing tolerance or thermal effects.

22. EMC, safety and test plan

Use suitable flanges, secure mechanical support and a continuous conducting path. Avoid exposed high-power microwave connections during operation. Follow transmitter power limits and applicable RF exposure procedures. For testing, calibrate the VNA using suitable waveguide standards, set reference planes at the flanges, measure S11 and S21 over 9.3–9.5 GHz, inspect flange condition and compare results with simulation. High-power testing should only use rated loads and appropriate safety procedures.

23. Final engineering decision

Initial design: NOT approved for final deployment until insertion loss, return loss, mode behaviour and power handling have been verified.

Improved design: suitable as a controlled academic prototype if the WR-90 assembly meets the specified band and measured acceptance targets. The illustrative 0.25 dB improved loss estimate is not a verified result. Final acceptance requires measured S-parameters and confirmation of actual hardware ratings.

If measurements fail, inspect flange mating, alignment and transitions; replace damaged or unsuitable sections; and compare measured response with a validated model. Final release must be based on measured performance rather than calculations alone.

References and Engineering Notes

1. Pozar, D. M., Microwave Engineering, 4th ed., John Wiley & Sons, 2012.

2. Collin, R. E., Foundations for Microwave Engineering, 2nd ed., IEEE Press, 2001.

3. Marcuvitz, N., Waveguide Handbook, MIT Radiation Laboratory Series, McGraw-Hill, 1951.

4. Balanis, C. A., Advanced Engineering Electromagnetics, 2nd ed., John Wiley & Sons, 2012.

5. Standard WR-90 waveguide dimensional data and manufacturer technical literature; verify current supplier revision for tolerances, material, finish and flange specifications.

6. Relevant IEEE/IEC microwave measurement guidance and instrument manufacturer's VNA calibration instructions.

Calculation summary

Quantity	Expression	Approximate result
Free-space wavelength	c/f at 9.4 GHz	31.9 mm
TE10 cutoff	c/(2a)	6.56 GHz
TE20 cutoff	c/a	13.11 GHz
TE01 cutoff	c/(2b)	14.75 GHz
Guide wavelength	$\lambda 0 / \sqrt{[1 - (fc/f)^2]}$	44.5 mm
Phase constant	$2\pi/\lambda g$	141 rad/m
Phase velocity	$c / \sqrt{[1 - (fc/f)^2]}$	4.18×10^8 m/s
Group velocity	$c \sqrt{[1 - (fc/f)^2]}$	2.15×10^8 m/s
TE impedance	$\eta 0 / \sqrt{[1 - (fc/f)^2]}$	526 Ω
1.5 m group delay	L/vg	6.98 ns

All calculated values are rounded. The calculation assumes an ideal air-filled WR-90 guide and should be checked against actual dimensions and operating frequency.

Appendix – MATLAB Starter Script

The MATLAB script calculates ideal first-order WR-90 propagation values across 9.3–9.5 GHz. It does not simulate conductor loss, flange discontinuities or full S-parameters.

```
clc; clear; close all;

c = 3e8;
a = 22.86e-3;          % WR-90 broad dimension (m)
b = 10.16e-3;          % WR-90 narrow dimension (m)
f = linspace(9.3e9, 9.5e9, 201);

fc10 = c/(2*a);
fc20 = c/a;
fc01 = c/(2*b);
lambda0 = c./f;
lambdag = lambda0 ./ sqrt(1 - (fc10./f).^2);
beta = 2*pi ./ lambdag;
vp = c ./ sqrt(1 - (fc10./f).^2);
vg = c .* sqrt(1 - (fc10./f).^2);
Zte = 376.730313 ./ sqrt(1 - (fc10./f).^2);
L = 1.5;
tau = L ./ vg;

fprintf('fc10 = %.3f GHz\\n', fc10/1e9);
fprintf('fc20 = %.3f GHz\\n', fc20/1e9);
fprintf('fc01 = %.3f GHz\\n', fc01/1e9);
idx = find(abs(f-9.4e9) == min(abs(f-9.4e9)),1);
fprintf('lambda_g = %.2f mm\\n', lambdag(idx)*1e3);
fprintf('beta = %.2f rad/m\\n', beta(idx));
fprintf('vp = %.3e m/s\\n', vp(idx));
fprintf('vg = %.3e m/s\\n', vg(idx));
fprintf('ZTE = %.1f ohm\\n', Zte(idx));
fprintf('1.5 m group delay = %.2f ns\\n', tau(idx)*1e9);

figure; plot(f/1e9,lambdag*1e3,'LineWidth',1.5);
xlabel('Frequency (GHz)'); ylabel('Guide wavelength (mm)');
title('WR-90 TE10 guide wavelength'); grid on;
figure; plot(f/1e9,beta,'LineWidth',1.5);
xlabel('Frequency (GHz)'); ylabel('Phase constant (rad/m)');
title('WR-90 TE10 phase constant'); grid on;
figure; plot(f/1e9,Zte,'LineWidth',1.5);
xlabel('Frequency (GHz)'); ylabel('TE wave impedance (ohm)');
title('WR-90 TE10 wave impedance'); grid on;
```

Appendix note

The script provides analytical propagation trends only. For actual insertion loss, return loss, flange effects, bends and transitions, use a full-wave electromagnetic solver or calibrated waveguide VNA measurements.